

ECON1050 Tutorial 9

Integration

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Today's Plan

- ▶ Recap: key ideas from Lecture 9
- ▶ Tutorial questions: Q1–Q4

Recap: Integration and Antiderivatives

Integration

indefinite
definite

Integration “undoes” differentiation. An **indefinite integral** is an antiderivative:

$$\int f(x) dx = F(x) + C, \quad F'(x) = f(x).$$

constant of integration

Constant of integration

C is an arbitrary constant. Extra information is needed to determine its value.

Recap: Rules for Indefinite Integrals

Power and constant rules

For $a \neq -1$,

$$\int \underline{x^a} \, dx = \frac{x^{a+1}}{a+1} + C, \quad \int k \, dx = kx + C.$$

Linearity

$$\begin{aligned} \int k f(x) \, dx &= k \int f(x) \, dx, \\ \int (f(x) \pm g(x)) \, dx &= \int f(x) \, dx \pm \int g(x) \, dx. \end{aligned}$$

Recap: Common Antiderivatives

Logarithmic forms

$$\int \frac{1}{x} dx = \ln |x| + C.$$

Exponential forms

$$\int e^x dx = e^x + C, \quad \int e^{kx} dx = \frac{1}{k} e^{kx} + C.$$

For $a > 0$, $a \neq 1$,

$$\int a^x dx = \frac{a^x}{\ln a} + C.$$

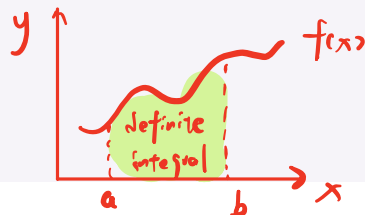
Recap: Definite integrals

Definition

A definite integral is a number:

$$\int_a^b f(x) dx.$$

It represents the **net area** under the curve between a and b .



Fundamental Theorem of Integral Calculus

If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

$F(b) - F(a)$

Net area vs total area

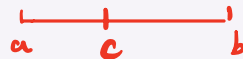
Areas above the x -axis count positively; areas below the x -axis count negatively.

Recap: Properties of definite integrals

Splitting an interval

For $a < c < b$,

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$



Changing limits

$$\int_a^b f(x) dx = - \int_b^a f(x) dx, \quad \int_a^a f(x) dx = 0.$$

Constant multiple

$$\int_a^b \alpha f(x) dx = \alpha \int_a^b f(x) dx.$$

Recap: Integration by Parts

Purpose

Integration by parts is used for products that cannot be integrated directly using the basic rules. It is the reverse of the product rule.

Formula

If f and g are differentiable,

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Handwritten notes above the formula:

- $f(x) \rightarrow f'(x)$
- $g'(x) \rightarrow g(x)$
- $f(x) \cdot g'(x)$ (above the integrand)
- $f(x) \cdot g(x) - \int f'(x) \cdot g(x) dx$ (above the right-hand side)

Equivalently,

$$\int u dv = uv - \int v du.$$

Recap: Integration by Substitution

Reverse chain rule

For an integral of the form

$$\int f(g(x))g'(x) \, dx,$$

let

$$u = g(x), \quad du = g'(x) \, dx.$$

$$u = g(x)$$

$$du = g'(x) \cdot dx$$

Then

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du = F(u) + C.$$

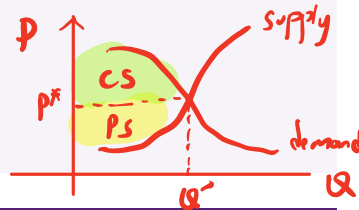
Finally substitute $u = g(x)$ back into the result.

Recap: Welfare Applications

Let inverse demand and supply be $P = D(Q)$ and $P = S(Q)$, with equilibrium (Q^*, P^*) .

Consumer surplus

$$CS = \int_0^{Q^*} (D(Q) - P^*) dQ = \int_0^{Q^*} \underline{D(Q)} dQ - P^* Q^*.$$



Producer surplus

$$PS = \int_0^{Q^*} (P^* - S(Q)) dQ = P^* Q^* - \int_0^{Q^*} S(Q) dQ.$$

Tutorial Q1

Question 1

Evaluate the following integrals:

(a) $\int (5x^3 + 2) dx,$

4 min

(b) $\int \left(2x^2 + \frac{1}{x^3} \right) dx,$

(c) $\int_1^3 4x(x^2 + 1) dx.$

$$(b) \int \left(2x^2 + \frac{1}{x^3} \right) dx = \int 2x^2 dx + \int \frac{1}{x^3} dx = 2 \int x^2 dx + \int x^{-3} dx = 2 \cdot \frac{x^3}{3} + C_1 + \frac{x^{-2}}{-2} + C_2 = \frac{2x^3}{3} - \frac{1}{2x^2} + C$$

$$(c) \int_1^3 4x(x^2 + 1) dx = \int_1^3 (4x^3 + 4x) dx = [x^4 + 2x^2]_{x=1}^{x=3} \\ = (3^4 + 2 \cdot 3^2) - (1^4 + 2 \cdot 1^2) \\ = 24$$

① antiderivative

② F.T.C

Tutorial Q1: Knowledge Used

Knowledge used

- ▶ Indefinite integrals and the constant of integration.
- ▶ Power rule and linearity of integration.
- ▶ Definite integrals and the Fundamental Theorem of Integral Calculus.

Tutorial Q1(a): Solution

Solution

Using linearity and the power rule,

$$\begin{aligned}\int (5x^3 + 2) \, dx &= \int 5x^3 \, dx + \int 2 \, dx \\ &= 5 \left(\frac{x^4}{4} \right) + 2x + C.\end{aligned}$$

Therefore,

$$\boxed{\int (5x^3 + 2) \, dx = \frac{5}{4}x^4 + 2x + C.}$$

Tutorial Q1(b): Solution

Solution

Rewrite $x^{-3} = 1/x^3$:

$$\begin{aligned}\int \left(2x^2 + \frac{1}{x^3} \right) dx &= 2 \int x^2 dx + \int x^{-3} dx \\ &= \frac{2}{3}x^3 - \frac{1}{2}x^{-2} + C.\end{aligned}$$

Hence,

$$\boxed{\int \left(2x^2 + \frac{1}{x^3} \right) dx = \frac{2}{3}x^3 - \frac{1}{2x^2} + C.}$$

Tutorial Q1(c): Solution

Solution

First expand the integrand:

$$4x(x^2 + 1) = 4x^3 + 4x.$$

Therefore,

$$\begin{aligned}\int_1^3 4x(x^2 + 1) \, dx &= \int_1^3 (4x^3 + 4x) \, dx \\ &= [x^4 + 2x^2]_1^3 \\ &= (81 + 18) - (1 + 2) \\ &= 96.\end{aligned}$$

Thus,

$$\boxed{\int_1^3 4x(x^2 + 1) \, dx = 96}.$$

Question 2

Use integration by parts to find:

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

(a) $\int \frac{\ln x}{x^2} dx, \quad x > 0,$

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = \frac{1}{x^2} \Rightarrow g(x) = -\frac{1}{x}$$

(b) $\int_1^2 4xe^{2x} dx.$

$$\begin{aligned} \text{(a)} \quad \int \frac{\ln x}{x^2} dx &= \int f(x) \cdot g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx \\ &= \ln x \cdot \left(-\frac{1}{x}\right) - \int \frac{1}{x} \cdot \left(-\frac{1}{x}\right) dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x} + C \end{aligned}$$

$$f(x) = 4x \Rightarrow f'(x) = 4$$

$$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2} \cdot e^{2x}$$

$$\begin{aligned} \text{(b)} \quad \int_1^2 4x \cdot e^{2x} dx &= \left[4x \cdot \frac{1}{2} \cdot e^{2x} \right]_{x=1}^{x=2} - \int_1^2 4 \cdot \frac{1}{2} \cdot e^{2x} dx \\ &= \left[2x \cdot e^{2x} \right]_{x=1}^{x=2} - \left[e^{2x} \right]_{x=1}^{x=2} \\ &= (2 \cdot 2 \cdot e^{2 \cdot 2} - 2 \cdot 1 \cdot e^2) - (e^{2 \cdot 2} - e^{2 \cdot 1}) = 3 \cdot e^4 - e^2 \end{aligned}$$

Tutorial Q2

Question 2

Use integration by parts to find:

5 min

(a) $\int \frac{\ln x}{x^2} dx, \quad x > 0,$

(b) $\int_1^2 4xe^{2x} dx.$

Formula

If f and g are differentiable,

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Equivalently,

$$\int u dv = uv - \int v du.$$

Tutorial Q2: Knowledge Used

Knowledge used

- Integration by parts:

$$\int f(x)g'(x) \, dx = f(x)g(x) - \int f'(x)g(x) \, dx.$$

- Antiderivatives of x^{-2} and e^{2x} .
- Evaluation of a definite integral using upper and lower limits.

Tutorial Q2(a): Solution

Solution

Let

$$f(x) = \ln x, \quad f'(x) = \frac{1}{x},$$
$$g'(x) = \frac{1}{x^2} = x^{-2}, \quad g(x) = \int x^{-2} dx = -\frac{1}{x}.$$

Then

$$\begin{aligned} \int \frac{\ln x}{x^2} dx &= \ln x \left(-\frac{1}{x} \right) - \int \frac{1}{x} \left(-\frac{1}{x} \right) dx \\ &= -\frac{\ln x}{x} + \int x^{-2} dx \\ &= -\frac{\ln x}{x} - \frac{1}{x} + C. \end{aligned}$$

Hence,

$$\boxed{\int \frac{\ln x}{x^2} dx = -\frac{\ln x + 1}{x} + C.}$$

Tutorial Q2(b): Solution

Solution

Let

$$f(x) = 4x, \quad f'(x) = 4, \quad g'(x) = e^{2x}, \quad g(x) = \frac{1}{2}e^{2x}.$$

Then

$$\begin{aligned} \int_1^2 4xe^{2x} dx &= \left[4x \left(\frac{1}{2}e^{2x} \right) \right]_1^2 - \int_1^2 4 \left(\frac{1}{2}e^{2x} \right) dx \\ &= \left[2xe^{2x} \right]_1^2 - 2 \int_1^2 e^{2x} dx \\ &= \left[2xe^{2x} - e^{2x} \right]_1^2 \\ &= 3e^4 - e^2. \end{aligned}$$

Therefore,

$$\boxed{\int_1^2 4xe^{2x} dx = 3e^4 - e^2 \approx 156.4}.$$

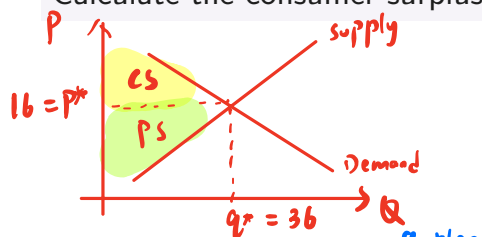
Tutorial Q3

Question 3

The demand and supply functions for a good are

$$q_D = 100 - 4p, \quad q_S = -12 + 3p.$$

Calculate the consumer surplus and the producer surplus at equilibrium.



$$\begin{cases} q = 100 - 4p & \text{①} \Leftrightarrow p = \frac{-q + 100}{4} \\ q = -12 + 3p & \text{②} \Leftrightarrow p = \frac{q + 12}{3} \end{cases}$$

$$\text{②} - \text{①} : 0 = (-12 + 3p) - (100 - 4p) \\ p^* = 16$$

$$\text{plug } p^* = 16 \text{ into ①: } q^* = 100 - 4 \cdot p^* = 36$$

$$\begin{aligned} CS &= \int_0^{q^*} \text{Demand } dq - p^* \cdot q^* \\ &= \int_0^{36} \frac{-q + 100}{4} \cdot dq - 16 \cdot 36 \\ &= \left[-\frac{1}{8}q^2 + 25q \right]_{x=0}^{x=36} - 576 = \left[-\frac{1}{8} \cdot 36^2 + 25 \cdot 36 - (0 + 0) \right] - 576 = 162 \end{aligned}$$

$$\begin{aligned} PS &= p^* \cdot q^* - \int_0^{q^*} \text{Supply } dq \\ &= 576 - \int_0^{36} \frac{q + 12}{3} dq \\ &= 576 - \left[\frac{q^2}{6} + 4q \right]_{x=0}^{x=36} \\ &= 576 - \left[4 \cdot 36 + \frac{1}{6} \cdot 36^2 - (0 + 0) \right] = 216 \end{aligned}$$

Tutorial Q3

Question 3

The demand and supply functions for a good are

5 mins

$$q_D = 100 - 4p, \quad q_S = -12 + 3p.$$

Calculate the consumer surplus and the producer surplus at equilibrium.

Consumer surplus

$$CS = \int_0^{Q^*} (D(Q) - P^*) dQ = \int_0^{Q^*} D(Q) dQ - P^* Q^*.$$

Producer surplus

$$PS = \int_0^{Q^*} (P^* - S(Q)) dQ = P^* Q^* - \int_0^{Q^*} S(Q) dQ.$$

Tutorial Q3: Knowledge Used

Knowledge used

- ▶ Market equilibrium: demand equals supply.
- ▶ Convert demand and supply into inverse functions $p_D(q)$ and $p_S(q)$.
- ▶ Consumer surplus and producer surplus as definite integrals.

Tutorial Q3: Equilibrium

Solution

From

$$q_D = 100 - 4p, \quad q_S = -12 + 3p,$$

the inverse demand and supply functions are

$$p_D(q) = 25 - \frac{1}{4}q, \quad p_S(q) = 4 + \frac{1}{3}q.$$

At equilibrium, $p_D(q) = p_S(q)$:

$$\begin{aligned} 25 - \frac{1}{4}q &= 4 + \frac{1}{3}q \\ \frac{7}{12}q &= 21. \end{aligned}$$

Hence,

$$\boxed{q^* = 36, \quad p^* = 16}.$$

Tutorial Q3: Consumer Surplus

Solution

Using

$$CS = \int_0^{q^*} p_D(q) \, dq - p^* q^*,$$

we have

$$\begin{aligned} CS &= \int_0^{36} \left(25 - \frac{1}{4}q \right) \, dq - 16(36) \\ &= \left[25q - \frac{1}{8}q^2 \right]_0^{36} - 576 \\ &= 25(36) - \frac{1}{8}(36)^2 - 576 \\ &= 162. \end{aligned}$$

Therefore,

$$\boxed{CS = 162}.$$

Tutorial Q3: Producer Surplus

Solution

Using

$$PS = p^* q^* - \int_0^{q^*} p_S(q) \, dq,$$

we have

$$\begin{aligned} PS &= 16(36) - \int_0^{36} \left(4 + \frac{1}{3}q \right) \, dq \\ &= 576 - \left[4q + \frac{1}{6}q^2 \right]_0^{36} \\ &= 576 - 4(36) - \frac{1}{6}(36)^2 \\ &= 216. \end{aligned}$$

Therefore,

$$\boxed{PS = 216}.$$

At equilibrium, consumer surplus is 162 and producer surplus is 216.

Tutorial Q4

Question 4

Use integration by parts to show that

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

$$\int \ln x dx = x \ln x - x + C.$$

3 marks

Hint: write $\ln x = 1 \cdot \ln x$.

$$\begin{aligned} \int \ln x dx &= \int 1 \cdot \ln x dx & \left\{ \begin{array}{l} f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x} \\ g'(x) = 1 \Rightarrow g(x) = x \end{array} \right. \\ &= \ln x \cdot x - \int \frac{1}{x} \cdot x dx \\ &= x \ln x - \int 1 \cdot dx \\ &= x \ln x - x + C \end{aligned}$$

Tutorial Q4

Question 4

Use integration by parts to show that

$$\int \ln x dx = x \ln x - x + C.$$

Hint: write $\ln x = 1 \cdot \ln x$.

Formula

If f and g are differentiable,

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Equivalently,

$$\int u dv = uv - \int v du.$$

Tutorial Q4: Knowledge Used

Knowledge used

- ▶ Integration by parts.
- ▶ Rewrite $\ln x$ as a product: $\ln x \cdot 1$.
- ▶ Derivative of $\ln x$ and antiderivative of 1 .

Tutorial Q4: Solution

Solution

Use

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Let

$$f(x) = \ln x, \quad f'(x) = \frac{1}{x}, \quad g'(x) = 1, \quad g(x) = x.$$

Then

$$\begin{aligned}\int \ln x dx &= x \ln x - \int \frac{1}{x} \cdot x dx \\ &= x \ln x - \int 1 dx \\ &= x \ln x - x + C.\end{aligned}$$

Therefore,

$$\boxed{\int \ln x dx = x \ln x - x + C.}$$

Key Takeaways

- ▶ Indefinite integration recovers antiderivatives; definite integration evaluates net accumulated area.
- ▶ The Fundamental Theorem links definite integrals to antiderivatives.
- ▶ Integration by parts reverses the product rule; substitution reverses the chain rule.
- ▶ Consumer and producer surplus can be calculated from areas represented by definite integrals.
- ▶ For Tutorial 9, the main techniques are the power rule, integration by parts, and welfare integrals.